generators are most powerful machines. They are used for large scale power generation.

According to Faraday's law of electromagnetic induction, an emf is induced in a circuit when magnetic flux linked with it changes. This emf is called induced emf. The magnitude of the induced emf is given by

$$\varepsilon = \frac{d\Phi_B}{dt} \quad or$$

$$\varepsilon = \frac{d}{dt}(BA\cos\theta) \qquad (4.17)$$

From the above equation, it is clear that induced emf can be produced by changing magnetic flux in any of the following ways.

(i) By changing the magnetic field B

(ii) By changing the area A of the coil and

(iii) By changing the relative orientation θ of the coil with magnetic field

4.4.2 Production of induced emf by changing the magnetic field

From Faraday's experiments on electromagnetic induction, it was discovered that an emf is induced in a circuit by changing the magnetic flux of the field through it. The change in flux is brought about by (i) relative motion between the circuit and the magnet (First experiment) (ii) variation in current flowing through the nearby coil (Second experiment).

4.4.3 Production of induced emf by changing the area of the coil

Consider a conducting rod of length l moving with a velocity v towards left on a rectangular fixed metallic framework as shown in Figure 4.23. The whole arrangement is placed in a uniform magnetic field $\vec{B}$ whose magnetic lines are perpendicularly directed into the plane of the paper.

As the rod moves from AB to DC in a time dt, the area enclosed by the loop and hence the magnetic flux through the loop decreases.

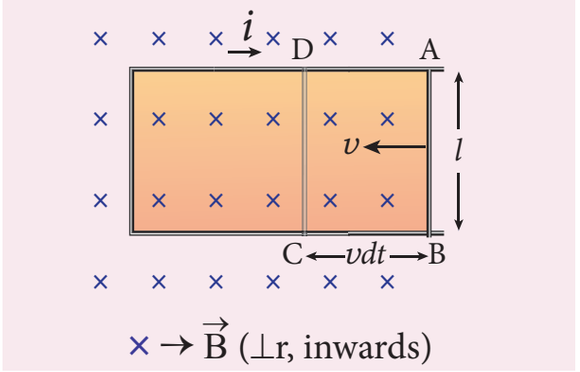


Figure 4.23 Production of induced emf by changing the area enclosed by the loop

The change in magnetic flux in time dt is

$$d\Phi_B = B \times \text{Change in area } (dA)$$

$$= B \times \text{Area } ABCD$$

$$\text{Since Area } ABCD = l(vdt)$$

$$d\Phi_B = Blvdt \quad \text{(or)}$$

$$\frac{d\Phi_B}{dt} = Blv$$

As a result of change in flux, an emf is generated in the loop. The magnitude of the induced emf is

$$\varepsilon = \frac{d\Phi_B}{dt}$$

$$\varepsilon = Blv \qquad (4.18)$$

This emf is known as **motional emf** since it is produced due to the movement of the conductor in the magnetic field. The direction of induced current is found to be clockwise from Fleming's right hand rule.

If R is the resistance of the loop, then the induced current is given by

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